SAKSHI PLATFORM

System Architecture & Verification Report

Evidence-Constrained Modus Operandi Reconstruction (EC-MORE)
Explainable AI Investigation Companion for Unnatural Death Investigations

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Platform Status: Phase 1 (100%) & Phase 2 (MOD-05/06 Complete)	**Test Suite:** 132 Tests Passing (0 Failures)
Framework: FastAPI + SQLAlchemy + SQLite/PostgreSQL	**Security Standard:** OIDC + Zero-Trust RBAC/ABAC

1. Executive Summary & Paradigm

SAKSHI is an explainable AI companion built to assist investigating officers (IOs) and supervisory authorities during complex unnatural death investigations. Under the **Evidence-Constrained Modus Operandi Reconstruction (EC-MORE)** paradigm, the platform enforces strict evidentiary grounding: no hypothesis or timeline can be constructed without verifiable, authenticated, and cryptographically anchored evidence.

The architecture establishes an immutable multi-phase pipeline where administrative governance, evidence ingestion, cryptographic security, and forensic reasoning remain cleanly decoupled.

Phase 1	Case Initiation & Governance	MOD-01 Case Management MOD-02 Identity / OIDC MOD-03 RBAC / ABAC MOD-04 Stage Engine	**100% COMPLETE** (92 tests)
Phase 2	Evidence Ingestion	MOD-05 Evidence Intake MOD-06 MIME / Signature Validation MOD-07 Antivirus Scan MOD-08 WORM Commit + Checksum	**ACTIVE** MOD-05: 100% MOD-06: 100% MOD-07/08: Next
Phase 3	Security & Integrity	Dual Hashing (SHA-256 + SHA3-512) WORM Object Lock & Merkle Audit Trail	Queued

2. Phase 1 — Case Initiation & Governance

Phase 1 anchors administrative accountability, identity verification, access authorization, and state machine lifecycle progression before any evidence can enter the system.

Module	Core Responsibilities	Security & Architectural Invariants
MOD-01 Case Management	• Mint collision-resistant Case IDs: SAKSHI-CASE-YYYY-HEX8 • Provision isolated Case Workspace boundaries • Bind FIR metadata & assigned Investigating Officer • Export Phase 2 Integration Contract	• Duplicate FIR prevention across station jurisdictions • Relational ID isolation (internal UUID separated from Case ID) • Immutable workspace directory anchoring

MOD-02 Identity / OIDC	• Centralized authentication boundary • JWT Bearer token validation (signature, issuer, expiry, audience) • Extraction of verified AuthenticatedInvestigator	• Zero credential storage in database • Rejection of forged client identity claims • Dual development mock & production OIDC modes
MOD-03 RBAC / ABAC	• Dual role & attribute policy evaluation • Dynamic permission registry (case:* and evidence:*) • Station/jurisdiction match checks • Explainable AuthorizationDecision with audit log	• **DEFAULT DENY** architecture • Client-supplied roles never trusted • Supervisory privilege escalation isolation
MOD-04 Stage Engine	• Deterministic case lifecycle state machine • Validates transitions: INITIALIZED -> EVIDENCE_INGESTION -> ... • Governs investigative feedback loops and closure	• Anti-skipping protection (no jumping to CLOSED/ANALYSIS) • Supervisory-only closure and reopening checks • Ingestion gate control

3. Phase 2 — Evidence Ingestion Pipeline

Phase 2 transforms raw multi-modal artifacts (autopsy reports, crime scene photos, 112 emergency calls, CDR spreadsheets, and digital forensics dumps) into registered, verified, and secured evidence objects.

Phase 2 Modular Pipeline Flow:

MOD-05: Evidence Intake
- Checks Stage Engine (EVIDENCE_INGESTION)
- Generates Evidence ID (SAKSHI-EVID-...)
- Sanitizes filenames (anti-traversal)
- Stages file in {workspace}/raw/
- Output: RegisteredEvidenceObject

→

MOD-06: MIME / Signature
- Reads bounded 8KB header
- Matches magic bytes vs declared type
- Rejects disguised executables/scripts
- Enforces 16-type allowlist
- Output: ValidatedEvidenceObject

→

MOD-07 & MOD-08 (Next)
- MOD-07: Antivirus scan
- MOD-08: WORM commit & pre-hash
- Secure handoff to Phase 3

MOD-05: Evidence Intake Specification

MOD-05 establishes forensic chain-of-custody at the moment of intake. It guarantees that files are associated with a valid Case ID, stored exclusively within the provisioned case workspace, and tied to the submitting officer. Filename sanitization strips null bytes, directory traversal delimiters (../, ..\), and filesystem escape sequences.

MOD-06: MIME / File Signature Validation Specification

MOD-06 enforces strict binary signature verification. It rejects the trust of client-provided file extensions and MIME headers. Key protections include:

- **Bounded 8KB Reads:** Inspects initial binary header bytes without loading large multi-gigabyte files into RAM.
- **Executable Trapping:** Detects Windows PE (MZ), Linux ELF (\x7fELF), Mach-O binaries, Java classes, and shell scripts (#!/bin/sh). Disguised executables are instantly REJECTED.
- **MIME Allowlist:** Enforces approved forensic formats across PDF, JPEG, PNG, TIFF, WEBP, WAV, MP3, OGG, FLAC, CSV, XLS, XLSX, ZIP, GZIP, and JSON.
- **Polyglot / Ambiguity Handling:** Reject files with conflicting or ambiguous binary signatures.

4. Data Models & Entity Contracts

Database Schema (`cases` and `evidence` tables)

```
cases table:
  - id (UUID PK)                    - case_id (Canonical Unique String, Indexed)
  - title (String)                  - fir_metadata (JSON structured metadata)
  - assigned_io_id (String)         - lifecycle_stage (INITIALIZED -> EVIDENCE_INGESTION...)
  - status (ACTIVE/CLOSED)          - case_workspace_path (Filesystem boundary)
  - created_by, created_at, updated_at

evidence table:
  - id (UUID PK)                    - evidence_id (SAKSHI-EVID-YYYY-HEX8, Unique)
  - case_id (FK -> cases.case_id) - original_file_name, stored_file_name
  - evidence_category (Enum)        - declared_mime_type (Client claimed)
  - detected_mime_type (Verified)   - detected_format (PDF, JPEG, WAV, CSV...)
  - file_size_bytes (Integer)       - status (REGISTERED_INTAKE, VALIDATED, REJECTED)
  - raw_storage_path (File path)    - submitting_investigator_id (Audit anchor)
  - validation_reason (Text)        - collection_location, collection_datetime
```

5. Security, Access Controls & Case Isolation

SAKSHI enforces defense-in-depth across authentication, authorization, and workspace boundary confinement:

Security Layer	Policy Implementation	Forensic Safeguard
Authentication (MOD-02)	Stateless JWT OIDC Tokens signed by trusted IdP. Extracts subject_id, roles, police station.	Prevents forged officer identities. Token expiration strictly enforced.
Authorization (MOD-03)	RBAC + ABAC matrix. Permissions: case:*, evidence:create, evidence:read, evidence:validate.	Default Deny. An officer from Station A cannot access or intake evidence into Station B cases without supervisory clearance.
Workspace Isolation	Resolved canonical paths verified against case_workspace_path.	Blocks path traversal (../../) and prevents cross-case tampering.
Audit Ledger	Structured JSON events logged for all transitions, intake, and validation verdicts via AuditIntegrationService.	Prepares append-only cryptographic Merkle audit trail for court admissibility.

6. Quality Assurance & Test Verification

The SAKSHI platform maintains 100% automated test coverage across unit and integration tiers. All previous Phase 1 tests remain non-regressed alongside new Phase 2 modules.

MOD-01	Case Management & Workspace Provisioning	16	100% (Passed)
MOD-02	Identity / OIDC Authentication Boundary	33	100% (Passed)
MOD-03	RBAC / ABAC Authorization Engine	25	100% (Passed)
MOD-04	Stage Engine Lifecycle Governance	18	100% (Passed)
MOD-05	Evidence Intake & Staging Service	15	100% (Passed)
MOD-06	MIME / File Signature Validation Service	25	100% (Passed)
TOTAL	**Entire SAKSHI Test Suite**	**132**	**100% (0 Failures)**

7. Phase 2 Roadmap & Next Steps

With MOD-05 (Intake) and MOD-06 (MIME Validation) 100% verified, the next implementation milestones in Phase 2 are:

- **MOD-07: Antivirus Scan:** Consumes ValidatedEvidenceObject to scan payloads for viruses, Trojans, and malware without executing files.
- **MOD-08: WORM Commit + Checksum:** Pre-hashes evidence payloads with SHA-256 and stages them for immutable Write-Once-Read-Many (WORM) storage.
- **Phase 3: Security & Integrity Handoff:** Dual hashing (SHA-256 + SHA3-512), AWS S3 Object Lock compliance, and append-only Merkle tree audit trails.

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